

ADF test

(1) type = "none"

$$\Delta y_t = \rho \cdot y_{t-1} + \epsilon_t$$

$$H_0: \rho = 0$$

$$H_1: \rho < 0 \equiv \rho \in (-2, 0)$$

(2) type = "drift"

$$\Delta y_t = \alpha_1 + \rho y_{t-1} + \epsilon_t$$

$$H_0: \rho = 0$$

$$H_1: \rho < 0 \equiv \rho \in (-2, 0)$$

(3) type = "trend"

$$\Delta y_t = \alpha_1 + \alpha_2 t + \rho y_{t-1} + \epsilon_t$$

$$H_0: \rho = 0$$

$$H_1: \rho < 0 \equiv \rho \in (-2, 0)$$

KPSS test

(1) type = "mu"

$$y_t = \mu + u_t$$

$$u_t = u_{t-1} + \epsilon_t$$

$$H_0: \sigma_\epsilon^2 = 0$$

$$H_1: \sigma_\epsilon^2 > 0$$

(2) type = "tau"

$$y_t = \mu + \tau \cdot t + u_t$$

$$u_t = u_{t-1} + \epsilon_t$$

$$H_0: \sigma_\epsilon^2 = 0$$

$$H_1: \sigma_\epsilon^2 > 0$$

PP test

(1) model = "constant"

~~y_t~~

$$y_t = \alpha + \rho \cdot y_{t-1} + \epsilon_t$$

$$H_0: \rho = 1$$

$$H_1: \rho \in (-1, 1) \\ \rho < 1$$

(2) model = "trend"

$$y_t = \alpha + \beta t + \rho \cdot y_{t-1} + \epsilon_t$$

$$H_0: \rho = 1$$

$$H_1: \rho < 1 \\ \rho \in (-1, 1)$$